Feasibility-Preserving Multi-Objective Harris Hawks Optimization for Employment-Based Visa Allocation

Javier Augusto Rebull Saucedo, Yazmín Ivonne Flores Martínez, and Raúl Gibrán Porras Alaniz

Universidad Autónoma de Ciudad Juárez,
Maestría en Inteligencia Artificial y Analítica de Datos,
Ciudad Juárez, Chihuahua, México
rebull@outlook.com

Abstract. Employment-based immigration to the United States allocates roughly 140,000 visas per fiscal year under a per-country ceiling and a categorical spillover cascade that together produce decade-long waits, large inter-country disparities, and thousands of unused visas. We cast annual allocation as a three-objective combinatorial program—minimizing unserved waiting load, inter-country disparity, and visa waste—under six hard policy constraints, and solve it with multi-objective Harris Hawks Optimization (MOHHO). The methodological core is a feasibility-preserving decoder: a smallest-position-value random-key mapping composed with a deterministic greedy assignment that satisfies all six constraints by construction, so the swarm never leaves the feasible region and needs no penalty terms or repair. We tune the population size and iteration budget with a Taguchi L9 orthogonal array and a confirmation run. Across 30 runs the recovered Pareto front strictly dominates the incumbent first-in, first-out policy on all three objectives. A controlled six-method ladder then isolates *what* drives performance: the decoder alone (random restart) already beats a real-coded NSGA-II, and—our central finding—the representation–operator match, not the metaheuristic family, governs the result. Matching operators to the representation lifts three distinct paradigms (genetic algorithm, decomposition, and swarm) above every random-key method; the Discrete-MOHHO we introduce is the most stable optimizer. The findings hold across five perturbed-demand instances and replicate on a structurally distinct multidimensional knapsack. The method delivers an auditable menu of allocation policies rather than a single recommendation.

Keywords: Multi-objective optimization · Harris Hawks Optimization · Constraint handling · Random-key decoder · Permutation encoding · Taguchi design of experiments · Resource allocation

1 Introduction

Each fiscal year the United States Congress authorizes approximately 140,000 employment-based (EB) immigrant visas, distributed across five categories (EB-1 to EB-5). Two structural rules govern the distribution: a *per-country ceiling* of 7 % of the annual total ($P_c \approx 25{,}620$ visas) that no single country may exceed, and a *spillover* mechanism that cascades unused visas across categories (EB-4/5 → EB-1 → EB-2 → EB-3). The incumbent system processes petitions strictly first-in, first-out (FIFO) by priority date. The interaction of these rules makes the incumbent policy dysfunctional: two countries concentrate more than 60 % of demand and accumulate waits of 10–13 years, while the clash between per-country and per-category ceilings leaves thousands of visas unused.

This matters beyond accounting. Behind every backlogged petition is a household whose relocation, employment, and family reunification hinge on a priority date, so the allocation rule has direct welfare consequences. The problem admits three legitimate, mutually conflicting goals: serve the longest-waiting petitions (*efficiency*), distribute opportunity evenly across nationalities (*equity*), and waste no visas (*utilization*). No single allocation optimizes all three; the policy-relevant object is the set of best compromises—a Pareto front.

Computing that front is hard for three reasons. First, the allocation is *combinatorial*: it assigns integer visa counts to 105 country×category groups under six simultaneous constraints, and its feasible set generalizes the multidimensional knapsack problem, which is NP-hard [11]. Second, the objectives are genuinely *conflicting*, so the solution is a front rather than a point. Third—and most often underestimated—the *hard constraints* are unforgiving: a continuous population metaheuristic naturally proposes infeasible candidates, and the usual remedies (penalty functions, aggressive repair) either require delicate tuning or distort the very search landscape the metaheuristic explores.

We address these challenges with a multi-objective adaptation of Harris Hawks Optimization (MOHHO) whose centerpiece is a *feasibility-preserving decoder*. Concretely, this paper makes five contributions: *(i)* a decoder that couples a smallest-position-value (SPV) random-key mapping with a deterministic greedy assignment, guaranteeing all six policy constraints *by construction*—without penalties or repair, and therefore without perturbing the search landscape; *(ii)* a multi-objective HHO that pairs this decoder with a crowding-pruned external archive and applies it to a real, constraint-heavy allocation problem rather than to synthetic benchmarks; *(iii)* a Taguchi L9 orthogonal-array tuning of the population size and iteration budget, replacing empirical parameter choices with a systematic experimental design validated by a confirmation run; *(iv)* a controlled six-method ladder that, by lifting three distinct paradigms (GA, decomposition, and a representation-matched *Discrete-MOHHO* we introduce) to within 0.4 % of one another once operators match the representation, shows that the feasibility-preserving decoder and the representation–operator match, not the choice of metaheuristic, govern performance—with a direct operator-order-preservation measurement that explains *why* (interpolating GA operators are near-identity on the decoded permutation); and *(v)* a rigorous empirical protocol (30 runs, omnibus Kruskal–Wallis and Friedman/Nemenyi tests with effect sizes, strict domination of FIFO, robustness across five perturbed-demand instances, and a structurally distinct second problem on which the core result replicates) that makes every claim reproducible.

2 Background and Related Work

Harris Hawks Optimization. HHO [12] is a population-based metaheuristic inspired by the cooperative hunting of Harris's hawks. Its defining feature is a continuous transition between exploration and exploitation governed by a single physical quantity, the prey's *escape energy* E, which decays with the iteration count; when $|E| \geq 1$ the hawks explore, and when $|E| < 1$ they exploit through soft or hard besieges, optionally with rapid Lévy-flight dives. HHO was introduced for single-objective continuous optimization and has since been extended to the multi-objective setting.

Multi-objective HHO. Kuşoğlu and Yüzgeç [14] adapt HHO to multiple objectives with an external archive and crowding-based leader selection, and Çerçi and Dönmez [3] combine elite non-dominated sorting with a grid index. More recent work folds elitist non-dominated sorting into HHO directly [13] and enhances its operators for both single- and multi-objective settings [5]. These variants are validated mostly on continuous benchmark functions; their application to a *constrained combinatorial* allocation problem, where feasibility—not just convergence—is the binding difficulty, is comparatively unexplored. Our work occupies that gap.

Bridging continuous search and combinatorial structure. The random-key representation of Bean [2] encodes a permutation as a real vector and recovers it by sorting—the SPV rule. Random keys let a continuous operator move freely in $\mathbb{R}^d$ while a decoder interprets each point as a feasible discrete solution. We exploit exactly this separation of concerns: the hawks search a smooth box-constrained space, and a decoder absorbs all combinatorial structure. The idea has matured into biased random-key genetic algorithms [15] and, most recently, a general random-key optimizer that pairs a problem-specific decoder with *any* metaheuristic [4]—the paradigm we instantiate here for HHO, and which our ladder probes directly by varying the metaheuristic while holding the decoder fixed.

Constraint handling in metaheuristics. Three families dominate (see Rahimi et al. [17] for a recent survey). *Penalty methods* fold constraint violations into the objective, but require problem-specific weights and blur the boundary between feasible and infeasible regions. *Repair operators* project infeasible candidates back into the feasible set, but aggressive repair can collapse diversity and overwrite the stochasticity that drives exploration. *Decoder (feasibility-preserving) approaches* guarantee feasibility by construction, leaving the search landscape intact. We adopt the third family: our greedy decoder makes every decoded candidate satisfy all six constraints exactly, so no penalty weight is ever tuned and no repair is ever applied.

Baseline and methodology. NSGA-II [8] is the canonical multi-objective evolutionary algorithm and our benchmark; such algorithms are the standard tooling for hard allocation and scheduling problems [22]. For parameter selection we use the Taguchi orthogonal-array design [16], a standard tool for tuning stochastic algorithms with few experiments and increasingly applied to metaheuristics [1]. We assess fronts with the hypervolume indicator [23] and with inverted generational distance and spacing [6], and we test significance with nonparametric procedures [9] and the A_{12} effect size [20].

3 Problem Formulation

Decision variables. The problem is discretized into $G = C \times |\mathcal{J}| = 21 \times 5 = 105$ *groups*, where each group g is a (country c, category j) pair. The decision vector is $\mathbf{x} = (x_1, \ldots, x_{105}),\ x_g \in \mathbb{Z}_0^+$, where x_g is the number of visas assigned to group g. Each group has a demand (backlog) n_g and a *waiting weight* $w_g = t_{\text{now}} - d_g$, the years elapsed since its priority date d_g.

Objectives. We minimize three conflicting objectives:

$$f_1(\mathbf{x}) = \frac{\sum_g (n_g - x_g)\, w_g}{\sum_g n_g} \qquad \text{unserved waiting load (efficiency),} \tag{1}$$

$$f_2(\mathbf{x}) = \max_{c_1, c_2 \in \mathcal{C}} \left| \bar{W}_{c_1} - \bar{W}_{c_2} \right| \qquad \text{maximum inter-country disparity (equity),} \tag{2}$$

$$f_3(\mathbf{x}) = V - \textstyle\sum_g x_g \qquad \text{wasted (unassigned) visas (utilization),} \tag{3}$$

where V is the total available visas and $\bar{W}_c = \big(\sum_{g:\, c(g)=c} x_g w_g\big) / \big(\sum_{g:\, c(g)=c} x_g\big)$ is the visa-weighted mean wait of country c. Thus f_2 measures the gap between the longest- and shortest-waiting nationalities: the smaller f_2, the more equitable the distribution.

Constraints. The feasible set $\mathcal{F}$ is defined by six hard constraints:

R1: $\sum_g x_g \le V = 140{,}000$	(total visas)
R2: $\sum_{g:\, c(g)=c} x_g \le P_c = 25{,}620\ \ \forall c$	(7 % per-country ceiling)
R3: $\sum_{g:\, j(g)=j} x_g \le K_j^{\text{eff}}\ \ \forall j$	(effective per-category cap)
R4: $0 \le x_g \le n_g\ \ \forall g$	(do not exceed demand)
R5: $x_g \in \mathbb{Z}_0^+\ \ \forall g$	(integrality)
R6: FIFO order within each (c, j) queue	(seniority within a group)

where K_j^{eff} is the per-category cap after the spillover cascade. The result is a three-objective integer program with six constraints whose feasible space grows combinatorially with G, motivating a metaheuristic rather than exact enumeration.

Table 1. The six MOHHO movement operators, selected by the escape energy $|E|$ and uniform draws q, r. The archive's leader supplies the prey position.

Op.	**Condition**	**Behavior**
OP1	$\|E\| \geq 1,\ q \geq 0.5$	Exploration from a random hawk in the swarm.
OP2	$\|E\| \geq 1,\ q < 0.5$	Exploration relative to the population centroid.
OP3	$0.5 \leq \|E\| < 1,\ r \geq 0.5$	Soft besiege: gradual approach to the prey.
OP4	$\|E\| < 0.5,\ r \geq 0.5$	Hard besiege: aggressive contraction.
OP5	$0.5 \leq \|E\| < 1,\ r < 0.5$	Soft besiege with rapid Lévy dives.
OP6	$\|E\| < 0.5,\ r < 0.5$	Hard besiege with rapid Lévy dives.

Pareto optimality. A solution $\mathbf{x}$ *dominates* $\mathbf{x}'$ ($\mathbf{x} \prec \mathbf{x}'$) iff $f_i(\mathbf{x}) \leq f_i(\mathbf{x}')$ for all i and $f_i(\mathbf{x}) < f_i(\mathbf{x}')$ for some i. The *Pareto front* is $\mathcal{P}^* = \{\mathbf{x} \in \mathcal{F} \mid \nexists \mathbf{x}' \in \mathcal{F} : \mathbf{x}' \prec \mathbf{x}\}$. MOHHO approximates $\mathcal{P}^*$ through an external archive.

4 Methods

4.1 Multi-objective Harris Hawks Optimization

Each hawk is a position $\mathbf{H} \in \mathbb{R}^{105}$ in the box $[0, 1]^{105}$. At iteration t the escape energy is

$$E = 2E_0\left(1 - \tfrac{t}{T}\right), \qquad E_0 \sim U(-1, 1), \tag{4}$$

which drives the choice among six movement operators selected by $|E|$ and uniform draws $q, r \in U(0, 1)$ (Table 1). The multi-objective adaptation introduces three elements. First, the *leader* ("rabbit") is not a single global best but a solution drawn from the external archive with probability proportional to its crowding distance, which biases search toward sparsely populated regions of the front. Second, an *external archive* stores non-dominated solutions and, when it overflows, is pruned by removing the solution with the smallest crowding distance. Third, the archive is updated after every move using the non-dominance relation. The archive size is treated as a tunable parameter (Section 5).

4.2 A feasibility-preserving decoder

The HHO operators act in the continuous space $\mathbb{R}^{105}$, but the problem is combinatorial with six hard constraints. We bridge the two with the SPV rule [2] followed by a deterministic greedy assignment (Algorithm 1). The SPV rule turns a hawk $\mathbf{H}$ into a permutation $\boldsymbol{\pi}$ by sorting the group indices in ascending order of their key h_g. The greedy decoder then walks the groups in that order and assigns

$$x_g = \min\left(n_g,\ V_{\text{rem}},\ P^{\text{rem}}_{c(g)},\ K^{\text{rem}}_{j(g)}\right), \tag{5}$$

decrementing the residual budgets of R1–R3 after each step. Because every assignment is capped by the remaining budgets and by demand, and counts are integers, *every* decoded vector satisfies R1–R6 by construction. No penalty term

Algorithm 1 Feasibility-preserving SPV + greedy decoder

Require: hawk $\mathbf{H} \in \mathbb{R}^{105}$; demands n_g; budgets V, $\{P_c\}$, $\{K_j^{\text{eff}}\}$
$\boldsymbol{\pi} \leftarrow \textsc{ArgSortAscending}(\mathbf{H})$ ▷ SPV: keys → permutation
$V_{\text{rem}} \leftarrow V$; $P_c^{\text{rem}} \leftarrow P_c\ \forall c$; $K_j^{\text{rem}} \leftarrow K_j^{\text{eff}}\ \forall j$
for $g \in \boldsymbol{\pi}$ **do** ▷ visit groups in SPV order
  $x_g \leftarrow \min\big(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}}\big)$
  $V_{\text{rem}} \mathrel{-}= x_g$; $P_{c(g)}^{\text{rem}} \mathrel{-}= x_g$; $K_{j(g)}^{\text{rem}} \mathrel{-}= x_g$
end for
return x ▷ satisfies R1–R6 without repair

is introduced and no repair is performed, so the objective landscape the hawks explore is exactly the feasible-region landscape—preserving both the structure of HHO's cooperative foraging and its stochasticity.

Proposition 1 (Feasibility by construction). *For any hawk* $\mathbf{H} \in \mathbb{R}^{105}$*, the allocation* $\mathbf{x}$ *returned by Algorithm 1 satisfies all six constraints R1–R6.*

Proof. The residuals are initialized to V, P_c, $K_j^{\text{eff}} \geq 0$ and, at each step, decremented by $x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$. By induction each residual stays ≥ 0, since x_g never exceeds the current residual; hence $0 \leq x_g \leq n_g$ (R4) and, as the minimum of nonnegative integers, $x_g \in \mathbb{Z}_0^+$ (R5). Because $x_g \leq V_{\text{rem}}$ and V_{rem} is the unconsumed total, $\sum_g x_g \leq V$ (R1); likewise $x_g \leq P_{c(g)}^{\text{rem}}$ and $x_g \leq K_{j(g)}^{\text{rem}}$ keep the per-country and per-category sums within P_c (R2) and K_j^{eff} (R3). Finally, each group is a single (c, j) pair sharing one priority date, so assigning the count x_g serves its x_g most-senior petitions, honoring FIFO order within the queue (R6). Thus $\mathbf{x} \in \mathcal{F}$ for every input, with no penalty term and no repair. □

4.3 Benchmarks and a representation-matched optimizer

To separate the contribution of the decoder, the representation, and the search heuristic, we compare a *ladder* of methods that all share the same greedy decoder, objectives, hypervolume reference point, and 50×500 budget; they differ only in how they explore. Two operate on the continuous SPV random keys in $\mathbb{R}^{105}$: *(a)* the MOHHO of Section 4.1; and *(b)* **NSGA-II** [8] with simulated binary crossover (SBX, $\eta_c = 20$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/d$), binary tournament selection, and elitist $(\mu+\lambda)$ replacement. A third, *(c)* **random restart**, draws the 25,000 candidates uniformly at random (no search), isolating what the decoder alone achieves.

The ablation (Section 6.4) reveals that the continuous random-key operators are a poor match for the induced permutation landscape. We therefore also evaluate three **permutation-native** methods—spanning three distinct paradigms—that act directly on permutations of the 105 groups (decoded by the same greedy procedure): *(d)* **permutation-NSGA-II** (dominance-based) with order crossover (OX) and swap mutation; *(e)* **permutation-MOEA/D** [21] (decomposition-based), with Tchebycheff scalarization over a simplex-lattice of weight vectors and the same OX + swap operators; and *(f)* **Discrete-MOHHO**, our representation-matched Harris Hawks optimizer. Discrete-MOHHO keeps HHO's signature—the escape energy E of Eq. (4) scheduling exploration and exploitation, and a crowding-pruned external archive supplying the leader—but expresses every move on permutations: when $|E| \geq 1$ a hawk explores by recombining (OX) with a random hawk or by a heavy random shuffle; when $|E| < 1$ it besieges the leader by inheriting an order segment from it (OX) and perturbing with a number of swaps that shrinks as $|E| \to 0$, with occasional Lévy-like segment reversals. All six methods run for 30 independent seeds under the identical budget.

4.4 Evaluation protocol and indicators

We assess each approximation front with four indicators: the *hypervolume* (HV; larger is better) dominated with respect to the reference point (10; 16; 50,000), an upper bound on each objective; the *inverted generational distance* (IGD; smaller is better) to a reference front Z formed by the non-dominated union of the MOHHO and NSGA-II combined fronts ($|Z| = 444$); the *spacing* indicator (smaller is better) measuring uniformity; and the *cardinality* of the non-dominated set. Objectives are min–max normalized before IGD and spacing. Each algorithm is run for 30 independent seeds, and a *combined front* is formed as the non-dominated union of its 30 per-run fronts. Statistical significance between the two 30-value HV distributions is assessed with a one-sided Mann–Whitney U test [9], and effect size with the Vargha–Delaney A_{12} [20]. The deterministic **FIFO** baseline orders groups by ascending priority date and is decoded with the same greedy procedure, yielding $f_1 = 7.2138$, $f_2 = 12.6377$ years, and $f_3 = 17{,}540$ wasted visas.

5 Parameter Tuning by Taguchi Orthogonal Array

Population-based metaheuristics are sensitive to their control parameters, and choosing them by trial and error is neither systematic nor reproducible. We therefore tune the two parameters that most directly govern the search effort—the population size N and the iteration budget T—together with two secondary parameters, the external-archive size and the Lévy exponent β, using a Taguchi orthogonal-array design [16]. The Taguchi method probes a four-factor space with a small, balanced set of experiments and quantifies each factor's influence through a signal-to-noise (S/N) ratio that rewards both a high mean response and low variance across replications.

Design. We use four factors at three levels each (Table 2). The response is the hypervolume HV of the final archive; because larger HV is better, we use the larger-the-better S/N ratio

$$\mathrm{S/N} = -10 \log_{10}\Big(\tfrac{1}{r} \textstyle\sum_{k=1}^{r} 1/y_k^2\Big), \tag{6}$$

Table 2. L9(3^4) orthogonal array, configurations, and response (12 replications per row). HV is reported as mean ± s.d. in units of 10^6; S/N is the larger-the-better ratio of Eq. (6).

Run	N	T	Archive	β	HV ($\times 10^6$)	S/N (dB)
1	30	100	50	1.3	1.744 ± 0.052	124.818
2	30	300	100	1.5	1.760 ± 0.043	124.904
3	30	500	150	1.7	1.763 ± 0.051	124.916
4	50	100	100	1.7	1.752 ± 0.043	124.862
5	50	300	150	1.3	1.803 ± 0.015	125.117
6	50	500	50	1.5	1.803 ± 0.012	125.121
7	70	100	150	1.5	1.781 ± 0.030	125.007
8	70	300	50	1.7	1.800 ± 0.013	125.104
9	70	500	100	1.3	1.809 ± 0.011	125.147

Table 3. Response table: mean S/N (dB) per factor level, range Δ, and rank. Larger Δ means a more influential factor; the highest-S/N level in each row is in bold (levels tied to rounding are both bolded).

Factor	Level 1	Level 2	Level 3	Δ	Rank
A: population N	124.879	125.034	**125.086**	0.207	1
B: iteration budget T	124.896	125.042	**125.061**	0.166	2
C: archive size	**125.014**	124.971	**125.014**	0.043	4
D: Lévy exponent β	**125.027**	125.011	124.961	0.067	3

where y_k is the HV of replication k. A full-factorial design would need $3^4 = 81$ configurations; the L9(3^4) orthogonal array estimates all four main effects with just nine. Each configuration is replicated with $r = 12$ independent seeds drawn disjointly from the 30-run benchmark seeds, so the tuning data never contaminate the evaluation of Section 6.

Main effects. Averaging the S/N ratio over the levels of each factor gives the response table (Table 3) and the main-effects plot (Figure 1). The ranking by S/N range is unambiguous: *population size* ($\Delta = 0.207$ dB) and *iteration budget* ($\Delta = 0.166$ dB) dominate, whereas the archive size ($\Delta = 0.043$ dB) and the Lévy exponent ($\Delta = 0.067$ dB) lie within the run-to-run noise. This is itself an informative result—it confirms that the two parameters flagged as needing justification are precisely the influential ones, and that the algorithm is robust to the secondary choices. Both influential effects also *saturate*: most of the population benefit is captured going from $N = 30$ to $N = 50$ (+0.155 of the 0.207 dB), and most of the iteration benefit going from $T = 100$ to $T = 300$ (+0.146 of the 0.166 dB), with diminishing returns thereafter. The same monotone-but-saturating shape is visible in Figure 1.

Optimum and confirmation. Selecting the best level of each factor yields the predicted optimum $N = 70$, $T = 500$, archive $= 50$, $\beta = 1.3$ (the two archive

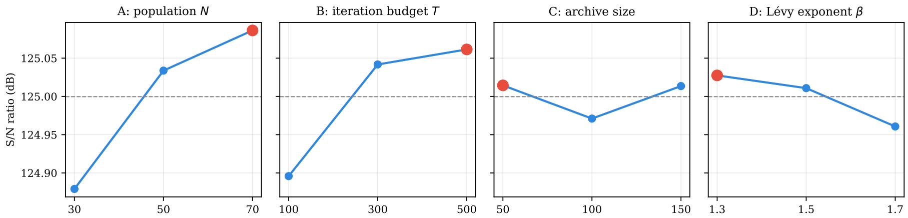

Fig. 1. Main-effects plot of the S/N ratio. The dashed line is the grand mean (125.00 dB) and the red marker the best level per factor. Population and iteration budget show clear, saturating trends; archive size and β are essentially flat.

levels 50 and 150 are tied to within rounding, so the archive choice is immaterial, consistent with its rank-4 influence). Under the additive model, its predicted S/N is ≈ 125.19 dB. A *confirmation run* at that configuration (12 fresh seeds) gives an observed S/N of 125.12 dB (HV = 1,803,110 $\pm$ 13,388). The ≈ 0.07 dB gap between prediction and observation is small—the slight positive bias is the expected signature of an additive model that ignores factor interactions—so the additive assumption holds and the tuning is internally consistent.

Adopted operating point. The 30-run study and the NSGA-II benchmark of Section 6 use $N = 50$, $T = 500$, archive = 100, and $\beta = 1.5$. A confirmation run at this configuration gives S/N = 125.078 dB (HV = 1,794,611 $\pm$ 20,553), within 0.47 % hypervolume of the L9 optimum. We retain it deliberately: the only *influential* factor in which it differs from the optimum is the population size, while the archive-size and β differences fall within the run-to-run noise; and increasing N from 50 to 70 raises the per-iteration cost by 40 % for a sub-0.5 % HV gain. Likewise $T = 500$ is intentionally conservative—it exceeds by a wide margin the empirical 99 %-convergence point at iteration 134 (Section 6.2)—so the budget cannot be blamed for any shortfall. In short, the Taguchi study turns a previously empirical choice into a justified operating point on the S/N plateau, where the influential factors are saturated and the rest are immaterial.

6 Results

6.1 Combined Pareto front and domination of FIFO

Combining and re-filtering the 30 runs yields a front of **406 non-dominated solutions**. Figure 2 shows it in the full objective space and Figure 3 in the f_1–f_2 projection; in both, the FIFO baseline floats *above* the front, visual evidence that it is dominated. The extreme solutions (Table 4) expose the trade-offs.

The central empirical finding is that **FIFO is not Pareto-optimal**: the minimum-f_1 solution improves f_1 from 7.2138 to 7.1857, f_2 from 12.64 to 10.74 years, and f_3 from 17,540 to 16,280 visas *simultaneously*. Moreover, 38 solutions of the front reach $f_3 = 0$ (zero waste), which FIFO never attains.

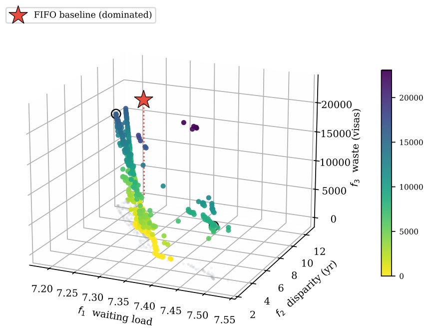

Fig. 2. The 406-solution combined Pareto front over the three objectives, colored by the waste f_3. The FIFO baseline (red star) floats above the front—the dotted drop-line marks its distance to the front's waste level—and is dominated; circles mark the extreme solutions.

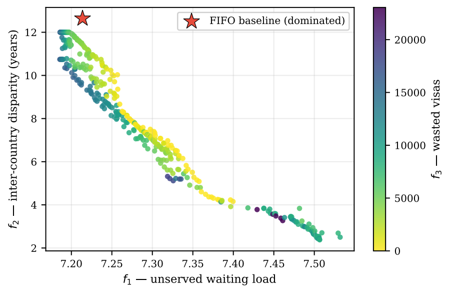

Fig. 3. f_1–f_2 projection of the same front, colored by f_3 (wasted visas). FIFO lies inside the dominated region.

Table 4. Extreme solutions of the combined front, compared with the FIFO baseline. The minimum-f_1 solution strictly dominates FIFO on all three objectives.

Solution	f_1	f_2 **(yr)**	f_3 **(visas)**	**Remark**
Min. f_1 (least waiting)	7.1857	10.7436	16,280	**Dominates FIFO**
Min. f_2 (most equitable)	7.5067	2.3792	8,664	Radical equity
Min. f_3 (full use)	7.2572	9.3611	0	100 % utilization
FIFO baseline	7.2138	12.6377	17,540	Reference (dominated)

6.2 Convergence and robustness

The mean hypervolume curve (Figure 4) reaches 95 % of its final value by iteration 9 and 99 % by iteration 134; the last 200 iterations contribute under 0.5 %. Over the 30 runs the HV distribution is narrow and symmetric (mean 1,785,803, standard deviation 27,831), a coefficient of variation of **1.56 %**, evidence of a stable and reproducible algorithm.

6.3 Comparison with NSGA-II

We first compare the two *random-key* optimizers under the identical problem, encoding, decoder, and budget of Section 5. MOHHO matches or outperforms the standard (real-coded) NSGA-II on every indicator, and *strictly* outperforms it on every aggregate front-quality measure (Table 5)—the two only tie on the two extreme values that the decoder makes attainable for both (best f_1 and zero waste). MOHHO attains a higher mean hypervolume (+7.2 %) with *lower* variability (CV 1.56 % vs. 2.84 %), converges markedly better (IGD 7.4× smaller), spreads

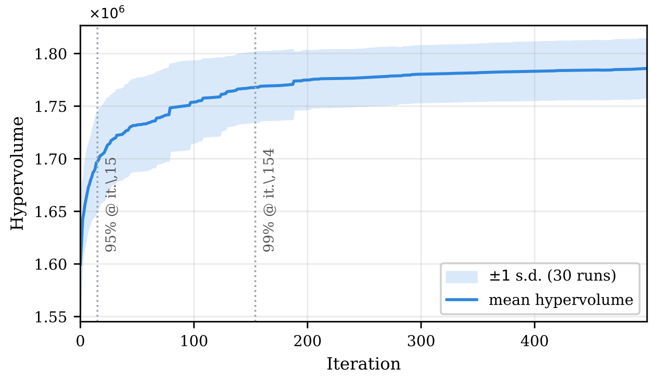

Fig. 4. Mean hypervolume over 30 runs with a ±1 standard-deviation band. The dotted markers indicate the iterations at which 95 % and 99 % of the final HV are attained; gains beyond iteration 134 are marginal.

solutions more uniformly, and recovers 2.3× more non-dominated solutions; the gap is significant (one-sided Mann–Whitney $p \approx 1.6 \times 10^{-10}$, $A_{12} = 0.97$). Figure 5 shows the front coverage. This advantage is a property of the *search dynamics*, not of archiving (granting NSGA-II the identical archive barely moves it; Section 6.4): MOHHO's position updates induce genuine permutation jumps on the random keys, whereas NSGA-II's interpolating operators scarcely perturb the decoded order (Section 6.4 quantifies this). That comparison, however, is only part of the story.

6.4 Decoder, representation, and search: a controlled ladder

Section 6.3 compared two random-key methods. To attribute performance to its true source we now climb a ladder of methods, all sharing the decoder and a 25,000 function-evaluation budget—the comparison currency (Table 6, Figure 6). The permutation-native methods inherit the Taguchi-tuned N and T and are not separately tuned, so any edge they show is not a tuning artifact.

The decoder does most of the work. Random restart—blind sampling with no search—already reaches a mean HV of 1,776,440, about 99.5 % of MOHHO's and, strikingly, 6.6 % *above* NSGA-II. The feasibility-preserving decoder shapes a landscape on which even random sampling is strong; classic MOHHO improves on it only marginally (+0.5 %; paired Wilcoxon $p = 0.028$, marginal under Bonferroni), and giving NSGA-II the *identical* archive barely helps it (1,671,109 vs. 1,666,143), so MOHHO's edge over NSGA-II is not an archiving artifact.

The representation–operator match is decisive. Why does a tuned NSGA-II fall below blind sampling? We measured the cause directly. On the SPV random

Table 5. MOHHO vs. NSGA-II under an identical problem, encoding, decoder, and budget (30 runs each). Best value per row in bold. The FIFO point is dominated by both and is not a front.

Metric	FIFO	MOHHO	NSGA-II
Method type	deterministic	metaheuristic	metaheuristic
Best f_1 (waiting load)	7.2138	7.1857	7.1857
Best f_2 (disparity, yr)	12.6377	**2.3792**	2.4817
Best f_3 (waste, visas)	17,540	0	0
Mean HV $\pm\,\sigma$ (30 runs)	—	**1,785,803 ± 27,831**	1,666,143 ± 47,289
Coefficient of variation	—	**1.56 %**	2.84 %
Combined-front HV	—	**1,831,654**	1,811,489
Non-dominated solutions	1	**406**	173
IGD vs. Z (smaller better)	—	**0.0033**	0.0245
Spacing (smaller better)	—	**0.0159**	0.0399
Significance vs. MOHHO	dominated	—	$p \approx 1.6 \times 10^{-10}$

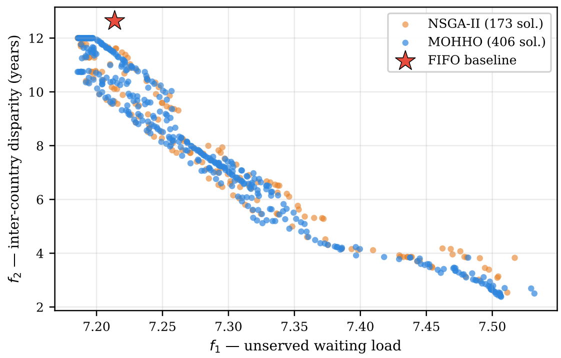

Fig. 5. Combined fronts of MOHHO and (real-coded) NSGA-II in the f_1–f_2 projection. MOHHO covers the front more densely and reaches lower-f_1 regions.

keys, SBX and polynomial mutation are *near-identity on the decoded permutation*: over 3,000 trials the Kendall rank correlation between a parent's SPV order and its child's is $\tau = 0.99$. The GA perturbs the keys so little that the induced permutation barely changes, leaving it almost frozen in combinatorial space—precisely why it sinks below blind sampling, which at least draws fresh permutations. HHO's position updates, by contrast, induce large permutation jumps ($\tau \approx 0$), so even on random keys the swarm genuinely traverses the decoded space (hence MOHHO > NSGA-II within the random-key tier). The clean fix, though, is to match the operators to the representation. Acting *directly* on permutations lifts *three different paradigms* above every random-key method: permutation-NSGA-II (dominance/GA) reaches 1,816,241, permutation-MOEA/D (decomposition) 1,814,179, and our **Discrete-MOHHO** (swarm) 1,808,850. The three cluster within 0.4 % of one another, yet all sit 1.3–1.7 %

above the best random-key method (classic MOHHO)—several times the spread *among* the matched paradigms. Discrete-MOHHO improves on classic MOHHO by +1.3 % (paired Wilcoxon $p = 5 \times 10^{-7}$, better on 25/30 seeds) and is the *most stable* method (CV 0.46 %, versus 0.61 % for permutation-NSGA-II and 1.56 % for classic MOHHO), making it the most dependable choice for a single-run decision. The marginal mean lead belongs to permutation-NSGA-II (+0.4 % over Discrete-MOHHO, $A_{12} = 0.26$).

An empirical regularity. The ladder (Figure 6) makes the pattern unmistakable: performance is governed by the *feasibility-preserving decoder* and the *representation–operator match*, not by the metaheuristic family. That encoding–operator mismatch can hurt search is known in principle since random keys were introduced [2]; our contribution is the *controlled, quantified* demonstration on a real constrained problem—including the counterintuitive result that a tuned NSGA-II falls below blind sampling—and the decomposition of performance into decoder, representation, and family. That three *distinct* paradigms (swarm, decomposition, dominance/GA) converge to within 0.4 % once the representation is matched, all far above the random-key methods, is strong evidence that the family is second-order here. We are precise about what this shows: the three matched paradigms share the OX + swap operators by design, so the dimension we vary—and find second-order—is the *selection/replacement architecture* (Pareto sorting, scalarized decomposition, or energy-driven besiege) given a representation-matched operator set; the operator and the representation themselves are *first-order*, as the random-key tier attests. The matched swarm reclaims competitiveness and the best stability, which is why we promote Discrete-MOHHO. An omnibus Kruskal–Wallis test across the six methods rejects equal performance decisively ($H = 136.4$, $p \approx 10^{-27}$), and the mean ranks split into two clean tiers—the three permutation-native methods (best) above the three random-key methods. The ranking is also robust to the hypervolume reference point (five reference points; ordering preserved throughout). We frame this as an empirical regularity for this problem, not a general law: all instances here share one structural skeleton—which the second, structurally distinct problem of Section 6.6 directly addresses.

6.5 Generalization across instances

To check that these findings are not specific to a single backlog estimate, we built five further instances by perturbing every group's demand n_g with an independent uniform ±20 % jitter (fixed seeds) and recomputing the spillover-adjusted category caps and the per-country caps accordingly; on each we re-ran the full comparison (10 seeds per method). The pattern is invariant (Figure 7): on all five perturbed instances—as on the base instance—the combined front strictly dominates FIFO, MOHHO outperforms NSGA-II (one-sided Mann–Whitney $p \leq 1.3 \times 10^{-4}$ in every case), and the decoder-dominance signature persists, with random restart reaching 97–99.5 % of MOHHO's hypervolume while NSGA-II trails at 91–93 %. The ordering MOHHO > random restart > NSGA-II therefore

Table 6. The method ladder: 30 runs each, identical greedy decoder and 25,000-evaluation budget. Top block: random-key encoding. Bottom block: permutation-native, spanning three distinct paradigms (swarm, decomposition, dominance/GA), which cluster within 0.4 % and dominate every random-key method. Best per column in bold; "perm-" abbreviates "permutation-".

Method	**Paradigm**	**Mean HV** $\pm\sigma$	**CV**	**Comb. HV**	**# sols**
NSGA-II	GA	1,666,143 ± 47,289	2.84 %	1,811,489	173
Random restart	—	1,776,440 ± 17,438	0.98 %	1,821,200	326
MOHHO	swarm	1,785,803 ± 27,831	1.56 %	1,831,654	**406**
Discrete-MOHHO	swarm	1,808,850 ± 8,264	**0.46 %**	1,839,510	396
perm-MOEA/D	decomp.	1,814,179 ± 8,396	0.46 %	1,841,569	366
perm-NSGA-II	GA	**1,816,241 ± 11,040**	0.61 %	**1,852,415**	372

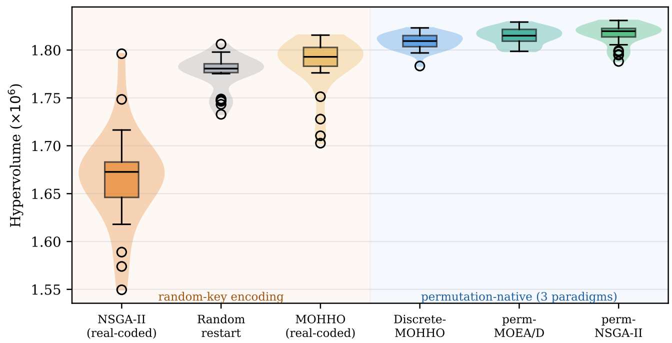

Fig. 6. The method ladder: per-run hypervolume (30 runs each, identical decoder and budget). Crossing from random-key encodings (left, shaded amber) to permutation-native operators (right, shaded blue) produces a clear jump; the metaheuristic family (GA vs. Harris Hawks) matters far less than the representation. Discrete-MOHHO is the tightest distribution (most stable).

holds across six instances, indicating that the conclusions are robust to ±20 % demand-estimation error rather than artifacts of one estimate. We are careful not to overstate this: all six instances share the same structural skeleton (21 countries, 5 categories, the same caps and the demand-far-exceeds-supply regime), so this is robustness to demand perturbation; generalization across structurally distinct problems is addressed directly in Section 6.6.

6.6 Generalization to a structurally distinct problem

The instances above share one structural skeleton. To probe whether the regularity reaches a genuinely different structure, we replicate the ladder on a classic, unrelated combinatorial problem: a three-objective multidimensional 0/1 knapsack (MOMKP; 120 items, four capacity constraints, three profit objectives) with *no*

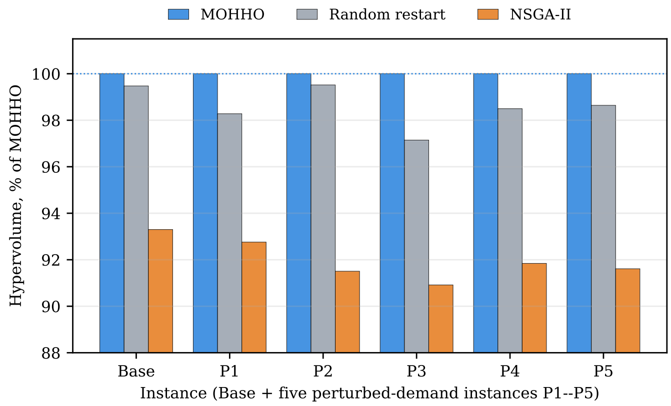

Fig. 7. Generalization over six instances (base plus five ±20 % perturbed-demand instances, P1–P5), each method's mean hypervolume shown as a percentage of MOHHO's on that instance. The ranking MOHHO > random restart > NSGA-II is preserved throughout, and FIFO is strictly dominated on every instance.

per-country caps, spillover, or waiting-time objective. Only the methodology is shared—a feasibility-preserving SPV + greedy decoder (add items in SPV order while every capacity holds) and the same six methods, budget, and 30 seeds. Because all methods here use identical seeds, we apply the paired Friedman test with a Nemenyi post-hoc (the Demšar standard).

The core finding replicates, the strict separation does not, and we report both (Figure 8; Friedman $\chi^2 = 128.6$, $p = 4.7 \times 10^{-26}$, critical difference 1.38). What *transfers*: the real-coded NSGA-II is again the *worst* method (mean rank 5.77), falling below random restart (5.23)—the counterintuitive "a tuned crossover-based GA underperforms blind sampling on random keys" result is not an artifact of the visa problem; and matching the GA to the representation again makes permutation-NSGA-II the best by a clear margin (rank 1.20; +32 % hypervolume over its real-coded form). What does *not* transfer: the clean random-key/permutation two-tier split. Here a real-coded Harris Hawks optimizer is unexpectedly strong (rank 2.67, statistically tied with Discrete-MOHHO at 2.57), so the real-coded swarm's *competitiveness is structure-specific*.

We can measure why. The operator-order analysis of Section 6.4 is a property of the encoding, hence problem-independent—we confirmed the operator τ on the MOMKP keys is unchanged (SBX 0.99, HHO $\approx$ 0, identical to the visa values)—so it accounts for the *robust* GA-freeze (real-coded GA worst on both problems) but not for the structure-specificity. That second phenomenon we leave, after testing, as a genuine open question. The natural hypothesis is that it tracks *search headroom*—how much the decoder leaves for any search to add (random restart is within 2.2 % of the best method on the visa problem but trails by 22.9 % on MOMKP). We tested this directly with a five-level sweep of the MOMKP capacity and it does *not* hold: within MOMKP the real-coded

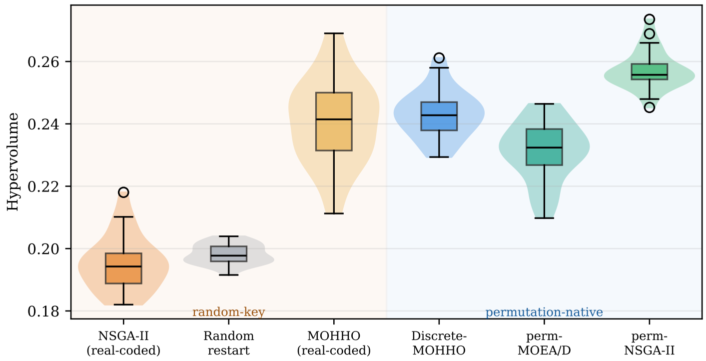

Fig. 8. The ladder on a structurally distinct problem (MOMKP). The real-coded NSGA-II and random restart remain at the bottom and permutation-NSGA-II at the top—as on the visa problem—but the real-coded Harris Hawks optimizer is competitive here, so the clean two-tier split is structure-specific. The robust transfer is the GA result, not the full ordering.

swarm's capture fraction is, if anything, *negatively* related to headroom (Spearman $\rho = -0.90$), and the visa point does not lie on the MOMKP trend, so a single headroom scalar does not unify the two problems. We therefore report structure-specificity as an open question rather than force a mechanism the data reject—a transparency we consider more valuable than a tidy but false story. What we *do* establish is robust: the representation–operator match dominates for *interpolating, crossover-based* search on both structures (the real-coded GA is worst on both, mechanized by τ=0.99), and the metaheuristic family is second-order *chiefly among representation-matched methods*. This honest scoping is more useful than a clean but fragile "law."

7 Discussion

The trade-off is structural. The conflict among objectives is not an artifact of the model. Two countries concentrate over 60 % of demand with 10–13-year waits; the 7 % ceiling forbids fully serving them, so reducing disparity (f_2) forces unserved load to remain in those countries ($f_1 \uparrow$), and distributing equitably leaves idle capacity in saturated categories ($f_3 \uparrow$). The three extreme solutions of Table 4 are mutually non-dominated—empirical proof of genuine three-way conflict.

Sensitivity. Two observations bound the result. First, with an archive of 500 (no pruning) the natural front holds about 211 solutions per run, yet hypervolume differs by under 0.03 % from the pruned archive of 100—crowding-distance pruning removes redundancy without loss. Second, the range of f_1 across the front (0.35) is about 28× narrower than that of f_2: because total demand (868,098) far exceeds the 140,000 visas, almost any permutation leaves a similar unserved load, so f_1

variability is constrained by the *problem structure*, not by the algorithm. We nonetheless keep f_1 as a distinct objective rather than collapsing to a two-objective (f_2, f_3) problem: it still discriminates the efficiency-extreme solutions (Table 4) and makes the efficiency–equity tension explicit—indeed it is the minimum-f_1 policy that strictly dominates FIFO, which a two-objective reduction would obscure.

What governs performance. The ladder of Section 6.4 yields an empirical regularity we expect to recur: when a hard-constrained combinatorial problem is solved through a feasibility-preserving decoder, performance is dominated by the decoder and by matching the search operators to the *induced* representation—not by the metaheuristic family. A tuned real-coded NSGA-II falls *below* blind sampling because its interpolating operators are near-identity on the decoded permutation ($\tau = 0.99$; Section 6.4) and so barely search the combinatorial space; switching to permutation-native operators lifts three distinct paradigms (a GA, a decomposition method, and a Harris Hawks optimizer) to the top, within 0.4 % of one another. The practical guidance is twofold: invest design effort in the decoder and the representation before the search heuristic; and when a single dependable run matters—as in decision support—prefer the most stable optimizer, here Discrete-MOHHO.

Policy implication. The front is a quantified map of compromises for a decision-maker, from prioritizing efficiency ($f_1 = 7.19$) to radical equity ($f_2 = 2.38$) or full utilization ($f_3 = 0$). Notably, reducing disparity from 12.64 to 2.38 years costs only about 4.1 % in f_1 relative to the FIFO baseline, indicating that large equity gains are attainable at modest efficiency cost—consistent with proposals to reform the per-country caps [7,10]. Each front solution maps to a concrete allocation. Figure 9 shows the per-country reallocation of one balanced, full-utilization policy ($f_3 = 0$, $f_2 = 6.65$ years) against FIFO: it keeps the two highest-demand countries at their 7 % ceiling (unchanged) and redirects the visas that FIFO leaves in the residual “Rest of World” pool toward currently underserved nationalities such as the Philippines, South Korea, Pakistan, and Taiwan—an interpretable equity intervention rather than an opaque score.

Limitations. Four caveats bound the conclusions. The model covers a single fiscal year and ignores intertemporal backlog dynamics; it uses 21 representative countries rather than the full roster; demand n_g is estimated from public aggregate data [18,19] (although Section 6.5 shows the comparison is stable under ±20 % demand perturbations); and it assumes every assigned visa is used (no post-allocation refusals). These limitations affect external validity, not the internal comparison among the six methods, which all run on identical data with an identical decoder and budget.

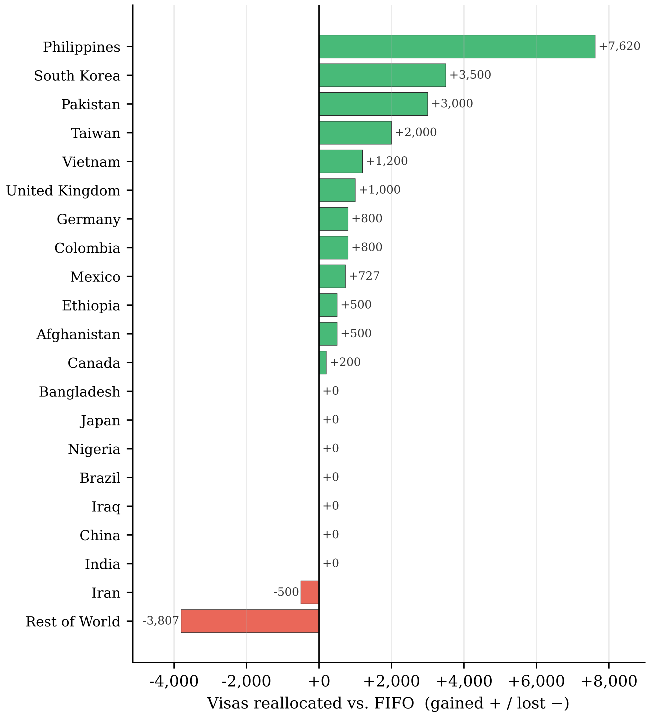

Fig. 9. Policy output: visas reallocated per country by a balanced full-utilization MOHHO solution relative to the FIFO baseline (gained in green, lost in red). The two highest-demand countries remain at their per-country ceiling; the redistribution is drawn mainly from the residual pool.

8 Conclusions

We formulated the annual allocation of U.S. employment-based visas as a three-objective integer program with six hard constraints and solved it through a feasibility-preserving SPV + greedy decoder that guarantees feasibility by construction—no penalties, no repair. A Taguchi L9 design replaced empirical parameter choices with a confirmed operating point. Across 30 independent runs the recovered Pareto front strictly dominates the FIFO incumbent on all three objectives, with 38 zero-waste solutions. Our central result comes from a controlled six-method ladder: the decoder alone (random restart) already beats a real-coded NSGA-II, and—the lesson—the representation–operator match, not the metaheuristic family, governs performance. Matching operators to the representation lifts three distinct paradigms (a GA, a decomposition method, and a Harris Hawks optimizer) to within 0.4 % of one another and above every random-key method; the Discrete-MOHHO we introduce improves on classic MOHHO by 1.3 % ($p = 5 \times 10^{-7}$) and is the most stable optimizer of all (CV

0.46 %), within 0.5 % of the best. All conclusions hold across five perturbed-demand instances, and on a structurally distinct multidimensional knapsack the core finding—a tuned real-coded GA falling below random sampling, reversed by representation-matched operators—replicates, while the two-tier split is structure-specific: operator order-preservation (problem-independent, τ=0.99) explains the robust GA-freeze, but the cause of the random-key swarm's structure-specific competitiveness resists a simple search-headroom explanation—which we test with a capacity sweep and reject—and we report it as an open question. Beyond the numbers, the method yields an auditable menu of allocation policies rather than a single recommendation. Future work includes a multi-period formulation, the full country roster, further problem domains beyond the two studied here, and additional optimizers such as SPEA2.

Data and Code Availability. The problem data, the optimizer, the Taguchi design, and the scripts that reproduce every figure and table are available at https://github.com/jrebull/MIAAD_Harrisv2.

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